Unit 5: Fundamentals of Solid-State Physics.

Basically, matters are broadly distinguishing into five different types Solid, liquid, gas, Plasma and condensed Bosen on the basis of their state. Further, the solid which has fixed macroscopic structure and are rigid in nature is divided as crystalline and Amorphous substances. The solid-state Physics deals with physical properties of solid basically crystal on the basis of their electronic conduction behaviour (thermal and electrical Conductivity).

- Crystalline Solid: These are the solid substances that is made up (composed) of a fixed geometrical shape and size structure. The regular and periodic repetition of such structure on three-dimensional space formed a crystalline Solid. Crystalline solid has high and sharp crystallization point, melting point and boiling point. They are anisotropic in nature i.e. the physical properties like conductivity, elasticity, tensile strength has different values with the different direction in solid. E.g. NaCl Crystal, ice, diamond, graphite, silicon, Quartz, most of metals cu, Al, Fe e.t.c.

by S.G. Sir

- Amorphous Solid: These are the solid substances that has rigid structure but have irregular geometrical shape and size structure is known as Amorphous solid. Amorphous solid has low melting point and boiling point. They are isotropic in nature i.e. the physical properties like conductivity, elasticity, tensile strength has same values with the different direction in solid. E.g. $CaCo_3$, Plastics, Sulphur, ordinary glass e.t.c.

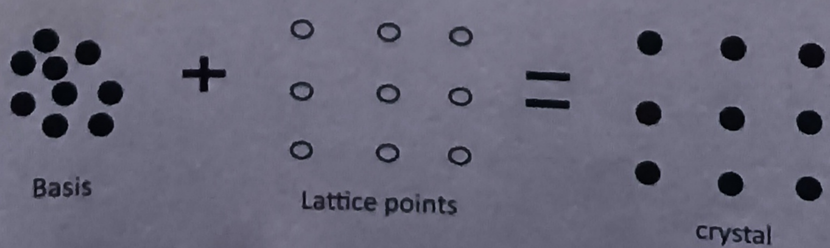
Crystal: A solid structure that has definite shape in which an atoms, molecules or ions are distributed in three-dimensional space with regular pattern is known as crystal. A crystal has a tendency to repeat itself in three dimensions infinitely. The periodic arrangement of atoms, molecules or ions in a crystal is called crystal lattice.

A crystal with an identical atom and are arrange in perfect regular distances is called ideal crystal. Practically, an ideal crystal doesn't exist in nature as there will some impurities atom or defect involves during the crystal birth.

A crystal lattice is composed of two parts which collectively formed a crystal i.e. basis and lattice points.

a. Basis: The group of atoms, molecules or ions in a crystal whose distribution in a spaced under definite design or position give rise a birth to crystal structure is known as Basis. Basis are the fundamental part of crystal that define characteristics features of it.

b. Lattice points: Lattice points are the imaginary points in three-dimensional space that are located at definite positions for definite crystal, the arrangement of basis over which forms a crystal. Lattice points define the shape and structure of particular crystal. The distance between two lattice points is called lattice parameter or lattice constant.



Unit cell: A small (minimum) volume cell or space of crystal lattice whose regular periodic repetition in three-dimensional space form a crystalline substance is known as unit cell. It is the fundamental structure or basic building block of crystal with definite shape, where the birth of crystal starts with this unit. Depending upon the dimension of crystal it may be two or three dimensional. Based on arrangement of basis (atoms or molecules or ions) in unit cell it is of two types;

a. Primitive Unit cell: A unit cell that contains lattice point or atoms present only at the corner of cell is called primitive unit cell.

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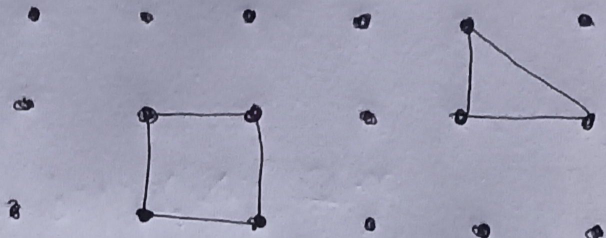


fig:- Primitive unit cell in 2D.

b. Non Primitive Unit cell: : A unit cell that contains lattice point or atoms present at the corner of cell along with other sites is called non primitive unit cell.

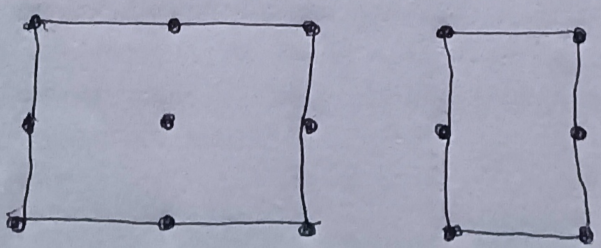


fig:- Non-primitive unit cell in 2D.

 # Types of Cubic unit cell in three-dimension:

A cubic unit cell has all its three lattice parameter equal to each other i.e. a = b = c and angles equal to 90^0 i.e. $\alpha = \beta = \gamma = 90^0$.

a. Simple cubic crystal (SCC): It is a three-dimensional primitive type cubic unit cell. It contains eight atoms at its each corner of cube. The atoms in this type of cube are just touching each other along the edge of cube as shown in fig,

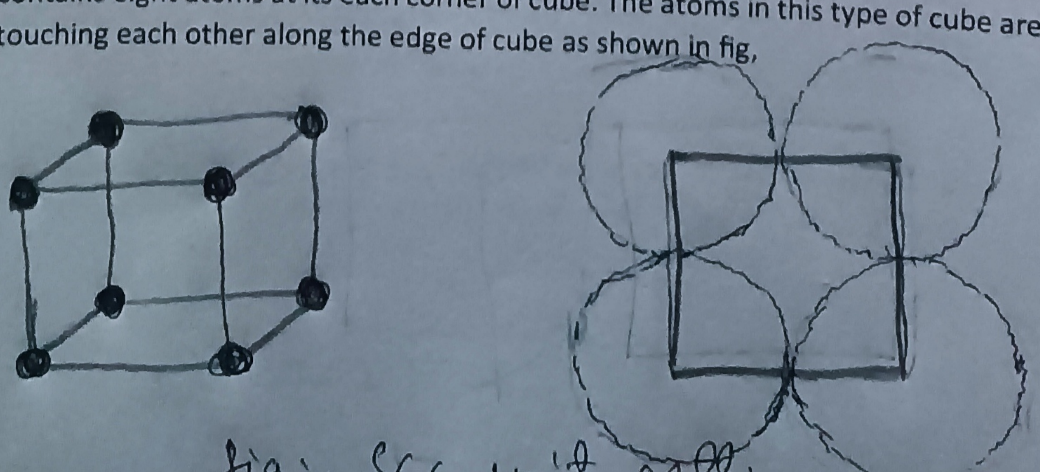
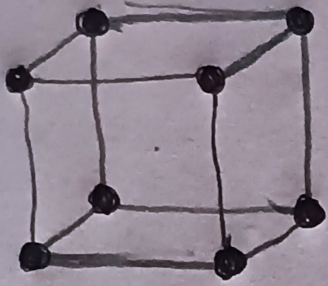


fig:- SCC unit cell.

b. Body centred cubic crystal (BCC): It is a three-dimensional non primitive type cubic unit cell. It contains eight atoms at its each corner of cube along with a central atom at the centre of body of cube. The atoms in cube are just touching each other along the body diagonal as shown in fig,



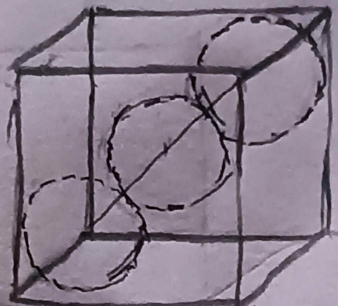


fig:- Bcc ~~[illegible]~~ unit cell.

c. Face centred cubic crystal (FCC): It is a three-dimensional non primitive type cubic unit cell. It contains eight atoms at its each corner of cube along with six central atoms at each centre of face of cube. The atoms in cube are just touching each other along the diagonal of face of cube.

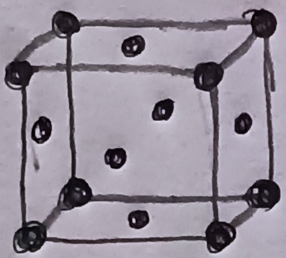
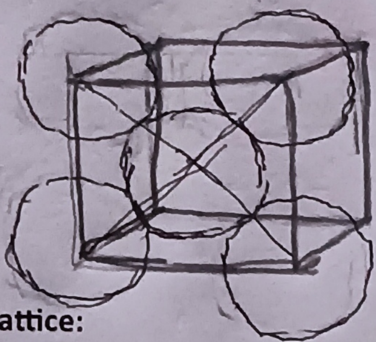


fig:- Fcc unit cell



 # Total number of atoms per unit cell of crystal lattice:

For a unit lattice cell having N_c, N_e, N_f, N_b numbers of atoms presents at corners, edge(side), face and inside the body of cube respectively then the total number of atoms present per unit cell of such crystal is given by,

$$N = \frac{N_c}{8} + \frac{N_e}{4} + \frac{N_f}{2} + N_b$$

by S. G Sir

Here, as the atom present at corner is share by eight-unit cell so it is counted as 1/8th part. Similarly, atom at edge of cube is share by four unit cell and is taken as 1/4th part, on so on, the atom present at face of cube is share by two unit cell so is counted as 1/2th part and atom present inside a unit cell belongs to only to that unit cell only (i.e.not share to others unit cell) and is counted as single.

Now, the total number of atoms present per unit cell of different cubic unit cell are

a. Simple cubic crystal:
Contains only eight corner atoms, one at each corner of cube.
$$N = \frac{8}{8} + \frac{0}{4} + \frac{0}{2} + 0 = 1$$

b. Body centered cubic crystal:
Contains eight corner atoms along with one atom inside a body of cube.
$$N = \frac{8}{8} + \frac{0}{4} + \frac{0}{2} + 1 = 2$$

c. Face centered cubic crystal: Contains eight corner atoms along with six face centered atoms at center of each face of cube.
$$N = \frac{8}{8} + \frac{0}{4} + \frac{6}{2} + 0 = 4$$

Atomic packing fraction (p.f) of crystal: An atomic packing fraction of a crystal is defined as the ratio of total volume occupied by the number of atoms present in unit cell to that of total volume of unit cell in space i.e.

$$p.f = \frac{\text{Total volume of atoms in unit cell}}{\text{Volume of unit cell}}$$

For, a crystal having 'N' number of atoms per unit cell then,

$$p.f = \frac{\text{Total number of atoms present per unit cell x volume of each atom in unit cell}}{\text{Volume of unit cell}}$$

It is a dimensional less constant since it is ration of same quantities. Its values is practically less than unit. The concept of atomic packing fraction gives an idea of the compactness of crystal which further related to different physical properties of crystal solid. Larger the value of p.f this implies more is the dense or compact ness of crystal, there is small vacant space or void is present inside it.

 # Determination of Atomic Packing fraction in case of different types of cubic unit cell:

a. **Simple cubic crystal (Scc) unit cell:** A simple cubic crystal unit cell is a primitive type of unit cell of cubic structure that contains eight corner atoms at each corner of cube. The corner atoms present in cube are identical and just touching each other along the side or edge of cube.

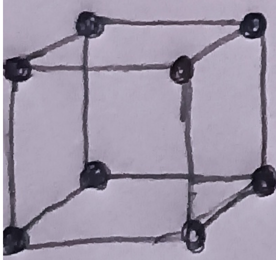
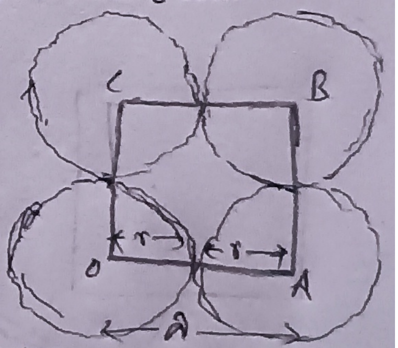


fig: Scc unit cell



So, the total number of atoms present in Scc unit cell is

$$N = \frac{8}{8} = 1$$

For, 'a' be the lattice parameter of cubic unit cell and 'r' be the radius of an atom present in it.

Then, from figure ; a=2r

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Now,

The atomic packing fraction of Scc unit cell

$$= \frac{\text{total number of atom per unit cell X volume of each atom}}{\text{volume of unit cell}}$$

$$= \frac{1 X \frac{4}{3}\pi r^3}{a^3}$$

$$= \frac{4\pi r^3}{3(2r)^3}$$

$$= \frac{\pi}{6}$$

$$= 0.52 \text{ or } 52\%$$

Thus, this shows that about only half of the volume of Scc unit cell is occupied by the atoms and hence, Scc unit cell is loosely packed cubic unit cell.

b. **Body centred cubic (bcc) unit cell:** A body centred cubic unit cell is a non- primitive type of unit cell of cubic structure that contains eight corner atoms at each corner of cube along with an atom at the center of cube inside it. The atoms present in cube are identical and corner atoms are just touching each other along the body diagonal of cube through body center atom.

So, the total number of atoms present in bcc unit cell is

$$N = \frac{8}{8} + 1 = 2$$

For, 'a' be the lattice parameter of cubic unit cell and 'r' be the radius of an atom present in it.

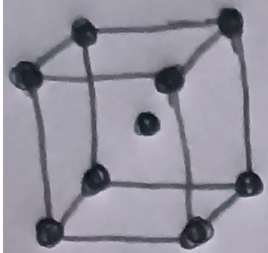

fig:- Bcc unit cell.

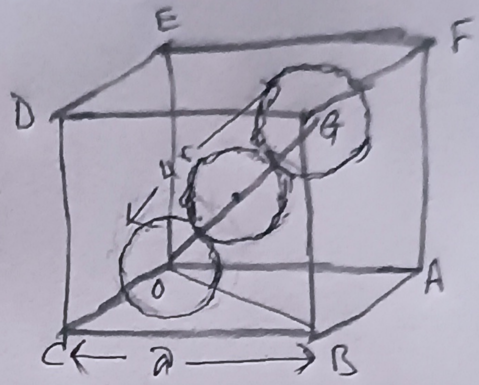

fig:- Atoms in Bcc unit cell

Then, from right angle triangle OGB,

$$OG^2 = OB^2 + BG^2$$

$$\text{Or, } (4r)^2 = OB^2 + a^2$$

$$\text{Or, } OB^2 = (4r)^2 - a^2$$

Again, from right angle triangle OBC,

$$OB^2 = OC^2 + CB^2$$

$$\text{Or, } (4r)^2 - a^2 = a^2 + a^2$$

$$\text{Or, } 16r^2 = 3a^2$$

$$\text{Or, } a^2 = 16r^2/3$$

$$\text{Or, } a = \frac{4}{\sqrt{3}} r$$

Now,

The atomic packing fraction of bcc unit cell is

$$= \frac{\text{total number of atom per unit cell X volume of each atom}}{\text{volume of unit cell}}$$

$$= \frac{2 X \frac{4}{3}\pi r^3}{a^3}$$

$$= \frac{8\pi r^3}{3(\frac{4}{\sqrt{3}} r)}$$

$$= \frac{8X\ 3\sqrt{3}\,\pi}{3X64}$$

$$= \frac{\sqrt{3}\,\pi}{8} a$$

$$= 0.68 \text{ or } 68\%$$

by S. G Sir

Thus, this shows that 68% of the volume of bcc unit cell is occupied by the atoms and only 32% of volume of unit cell is vacant. In comparison to Scc unit cell, bcc cubic unit cell is tightly packed in nature.

c. **Face centred cubic (Fcc) unit cell:** A face centred cubic unit cell is a non- primitive type of unit cell of cubic structure that contains eight corner atoms at each corner of cube along with six atoms at the center of the each face of cube. The atoms present in cube are identical and corner atoms are just touching each other along the face diagonal through face centred atom of cube.

So, the total number of atoms present in bcc unit cell is

$$N = \frac{8}{8} + \frac{6}{2} = 4$$

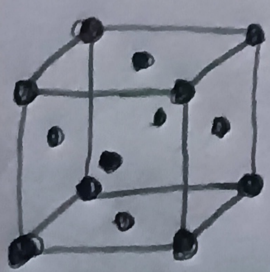

fig:- Fcc unit cell.

For, 'a' be the lattice parameter of cubic unit cell and 'r' be the radius of an atom present in it.

Then, from right angle triangle ABD,

$$DB^2 = AB^2 + AD^2$$

$$\text{Or, } (4r)^2 = a^2 + a^2$$

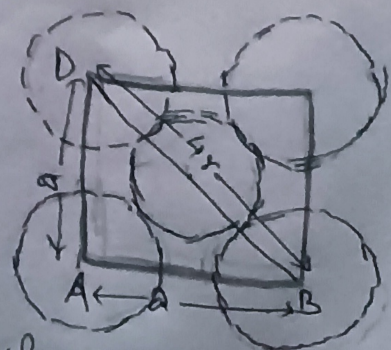

fig:- Atoms in fcc unit cell.

Or, $16r^2 = 2a^2$

Or, $8r^2 = a^2$

Or, $a = 2\sqrt{2}\,r$

Now, The atomic packing fraction of Fcc unit cell is

$$= \frac{\text{total number of atom per unit cell X volume of each atom}}{\text{volume of unit cell}}$$

$$= \frac{4\,X\,\frac{4}{3}\pi r^3}{a^3}$$

$$= \frac{16\,\pi\,r^3}{3(2\sqrt{2}\,r)^3}$$

$$= \frac{16X\pi}{3X\,16\,\sqrt{2}}$$

$$= \frac{\pi}{3\sqrt{2}}\,\text{a}$$

$$= 0.74 \text{ or } 74\%$$

This shows that 74% of the volume of Fcc unit cell is occupied by the atoms and only 26% of volume of unit cell is vacant. Fcc cubic unit cell is more tightly packed (highly dense) crystal in nature.

 # Expression of density of cubic unit cell in terms its number of atoms per unit cell :

Let, us consider a cubic unit cell having 'ρ' density , 'a' lattice parameter and containing 'N' number of atoms or molecules per unit cell.

Then, density of unit cell (ρ)= $\frac{\text{mass of unit cell}}{\text{volume of unit cell}} = \frac{m}{a^3}$ 1

where m is total mass of unit cell.

Similarly, For 'M' be molecular weight of molecule present in unit cell;
using 1 mole concept,

'N_A' Avogardro's number of atoms = M molar mass

Or, Mass of 1 atom = $\frac{M}{N_A}$

Or, Mass of N atom = $N\frac{M}{N_A}$ by S.G Sir

Or, mass of unit cell (m) = $N\frac{M}{N_A}$

Now, from equation 1;

density of unit cell (ρ) = $\frac{NM}{N_A\,a^3}$

This is required relationship between density of cubic unit cell with its lattice parameter and number of atoms per unit cell.

Further,

density of unit cell (ρ)= $\frac{nM}{N_A}$;

where $n = \frac{N}{a^3}$ is a concentration of atom or molecules in unit cell i.e number of atoms or molecules per unit volume of unit cell.